

Homework IoT - Exercise 1

Mattia Brianti 10773859
Alex Hathaway 10829946

Consider the following pseudocode for an ESP32-based vibration monitoring node:

```
1 // Global timer handle
2 declare timer_handle as esp_timer_handle_t
3
4 function setup_accelerometer():
5     initialize_accelerometer(sampling_rate = 100 Hz)
6
7 function setup_timer():
8     create_periodic_timer(callback = process_window,
9                             period = 10 seconds)
10
11 function process_window(arg):
12     vibration_peaks = count_vibration_peaks(last_10_seconds)
13
14     if vibration_peaks == 0:
15         payload = create_message(size = 512 bytes)
16     else if vibration_peaks == 1:
17         payload = create_message(size = 2 KB)
18     else:
19         payload = create_message(size = 4 KB)
20
21     send_payload(payload)
22
23 function app_main():
24     setup_accelerometer()
25     setup_timer()
26
27     loop forever:
28         delay(100 ms)
```

Assume that the number of vibration peaks detected in each 10-second window follows a Poisson distribution with an average rate:

$$\lambda = 0.4 \text{ peaks/window} \quad (1)$$

The system is operated using IEEE 802.15.4 in beacon-enabled mode, utilizing the CFP only. Assume:

- $R = 250 \text{ kbps}$
- $L = 128 \text{ bytes}$

Scenario Constraints

Assume that:

- one packet fits exactly into a single CFP slot.
- the CFP is dimensioned according to the expected traffic load.

- the reporting period is set to 10 seconds.
- the monitoring system is composed of 1 PAN coordinator and 3 vibration monitoring nodes

1 Question 1

Compute the Probability Mass Function of the output rate: $P(r=r_0), P(r=r_1), P(r=r_2)$ where r_0, r_1 , and r_2 are the output rates when the node detects 0, 1, or more than 1 vibration peaks, respectively.

Since the number of vibration peaks in a 10-second window follows a Poisson distribution with $\lambda = 0.4$ we can use this formula

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!} \quad (2)$$

to calculate the probabilities for each peak.

For 0 peaks ($k = 0$):

$$P(X = 0) = \frac{e^{-0.4} \cdot 0.4^0}{0!} \approx 0.6703$$

For 1 peak ($k = 1$):

$$P(X = 1) = \frac{e^{-0.4} \cdot 0.4^1}{1!} \approx 0.2681$$

For more than 1 peak ($k > 1$): since the sum of all probabilities must equal 1, we can subtract the probabilities of 0 and 1 from 1:

$$\begin{aligned} P(X > 1) &= 1 - (P(X = 0) + P(X = 1)) \\ &= 1 - (0.6703 + 0.2681) \approx 0.0616 \end{aligned}$$

Now we calculate the output rate, which is defined by the size of the payload generated over the reporting period. We treat 1 KB as 1000 bytes. The results are the following:

- Rate r_0 (0 peaks):
The payload size is 512 bytes.

$$r_0 = \frac{512 \text{ bytes} \cdot 8 \text{ bits/byte}}{10 \text{ seconds}} = 409.6 \text{ bps}$$

- Rate r_1 (1 peak):
The payload size is 2 KB, which is 2000 bytes.

$$r_1 = \frac{2000 \text{ bytes} \cdot 8 \text{ bits/byte}}{10 \text{ seconds}} = 1600.0 \text{ bps}$$

- Rate r_2 (> 1 peaks):
The payload size is 4 KB, which is 4000 bytes.

$$r_2 = \frac{4000 \text{ bytes} \cdot 8 \text{ bits/byte}}{10 \text{ seconds}} = 3200.0 \text{ bps}$$

Putting it all together, the Probability Mass Function of the output rate is:

- $P(r = 409.6 \text{ bps}) = 0.6703$
- $P(r = 1600 \text{ bps}) = 0.2681$
- $P(r = 3200.0 \text{ bps}) = 0.0616$

2 Question 2

Based on the output rate PMF, compute a consistent CFP slot assignment for the system.

To determine exactly how many slots in the CFP must be assigned to each of the 3 vibration monitoring nodes we divide the payload size by the packet size L.

- 0 Peaks (512 bytes):

$$\frac{512 \text{ bytes}}{128 \text{ bytes/slot}} = 4 \text{ slots}$$

- 1 Peak (2000 bytes):

$$\frac{2000 \text{ bytes}}{128 \text{ bytes/slot}} = 15.625 \text{ slots}$$

- >1 Peaks (4000 bytes):

$$\frac{4000 \text{ bytes}}{128 \text{ bytes/slot}} = 31.25 \text{ slots}$$

Of course since a node cannot transmit in a fraction of a slot, we must round up to 16 slots for 1 peak and 32 slots for more than 1 peak. To determine the correct number of slots we choose the worst case scenario (detecting more than 1 peak) which generates a 4000 byte payload. Therefore we must assign to each node 32 slots per Beacon Interval to guarantee it can always transmit data. Since the monitoring system is composed of 3 vibration monitoring nodes the total will add to 96 slots.

3 Question 3

Compute:

- T_s
- N_{slots} in the CFP
- T_{active}
- T_{inactive}
- The duty cycle of the system

3.1 T_s

T_s is found by dividing the number of bits in a packet by the nominal bit rate R.

$$T_s = \frac{L}{R} = \frac{(128 \times 8) \text{ bits}}{250,000 \text{ bps}} = 0.004096 \text{ seconds}$$

$$T_s = 4.096 \text{ ms}$$

3.2 N_{slots} in the CFP

This is calculated in point 2 and it's equal to 96 slots.

3.3 T_{active}

The active part of the superframe includes all the slots in the CFP plus one additional slot allocated for the beacon, which brings the total number of active slots to 97. To get the duration of the active part we multiply that with the duration of a single slot (T_s):

$$T_{\text{active}} = 97 \text{ slots} \cdot 4.096 \text{ ms/slot}$$

$$T_{\text{active}} = 397.312 \text{ ms}$$

3.4 T_{inactive}

The duration of the inactive part is the total Beacon Interval minus the active duration (T_{active}). The reporting period acts as our BI, which is 10 seconds.

$$T_{\text{inactive}} = BI - T_{\text{active}}$$

$$T_{\text{inactive}} = 10,000 \text{ ms} - 397.312 \text{ ms}$$

$$T_{\text{inactive}} = 9602.688 \text{ ms}$$

3.5 Duty cycle of the system

The duty cycle is the ratio of the active duration to the total Beacon Interval.

$$\eta = \frac{T_{\text{active}}}{BI}$$

$$\eta = \frac{397.312 \text{ ms}}{10,000 \text{ ms}} = 0.0397312$$

Duty Cycle (η) $\approx 3.97\%$

4 Question 4

How many additional vibration monitoring nodes can be added while keeping the duty cycle below 10%?

By multiplying the maximum possible duty cycle value by the total interval (BI), we can find the maximum active time tolerated by the system:

$$T_{\text{active,max}} = 0.10 \cdot BI = 0.10 \cdot 10000 \text{ ms} = 1000 \text{ ms}$$

After calculating that, we must convert it into the maximum number of equivalent slots. Knowing that the duration of a single slot (T_s) is 4.096 ms, we divide the maximum active time by it and round down, since as we said before we cannot transmit in a fraction slot:

$$N_{\text{active,max}} = \lfloor \frac{T_{\text{active,max}}}{T_s} \rfloor = \lfloor \frac{1000 \text{ ms}}{4.096 \text{ ms/slot}} \rfloor = 244 \text{ slots}$$

The result means that the system can have a maximum of 244 active slots to respect the 10% constraint. Now to calculate the available slots for the new nodes we subtract the ones already occupied (1 for sending the beacon and 96 occupied by the already present 3 nodes). In this case the number of remaining slots is $244 - 97 = 147$ slots. As we calculate it before, we know that each new node will require in the worst case scenario 32 slots to guarantee transmission. This means that to calculate the number of additional nodes we have to divide the remaining slots by the slots needed by a single node, which is:

$$\text{Additional nodes} = \lfloor \frac{147 \text{ slots}}{32 \text{ slots/node}} \rfloor = \lfloor 4.59 \rfloor = 4 \text{ nodes}$$